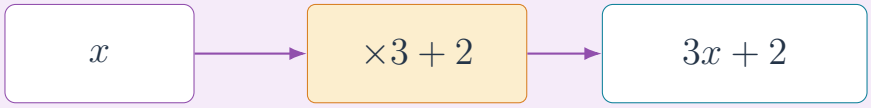
GANITA PRAKASH • CHAPTER 4

Mathematics | Part I

EXPRESSIONS USING LETTER-NUMBERS

Questions first → clear explanations → revision



Letters let us describe a pattern for any number

Learn it. Draw it. Explain it. Solve it.

Editable Class 7 notes aligned to the provided NCERT chapter

Contents

1 NCERT textbook questions: answers first **2**
1.1 Figure it Out — NCERT pp. 84–85 . . . 2
1.2 Figure it Out — NCERT pp. 93–94 . . . 3
1.3 Figure it Out — NCERT pp. 102–105 . . . 3

2 In-text examples, puzzles and discussion checks **6**

3 Student notes: learn the ideas **6**
3.1 What do letters mean? . . . 6
3.2 Visualise a growing pattern . . . 7
3.3 Brackets, like terms and signs . . . 7
3.4 Model a story with a formula . . . 7

4 Short extra practice with solutions **7**

5 One-page revision checklist **7**

NCERT textbook questions: answers first

Roadmap of the textbook questions

The chapter has three sets of *Figure it Out* exercises: **pp. 84–85 (7 questions), pp. 93–94 (picture sums, simplification and error corrections), and pp. 102–105 (15 questions)**. Each set is answered here. Open investigations and selected in-text puzzles follow the solved exercises.

Figure it Out — NCERT pp. 84–85

Set A • Q1: perimeter formulas

Question. Write formulas for an equal-sided triangle, a regular pentagon and a regular hexagon.

Answer. If each side measures a, the formulas are **(a)** $3a$, **(b)** $5a$, **(c)** $6a$. Reason: perimeter is the sum of all side lengths.

Set A • Q2: two pipes

Question. One pipe is 20 m long and the second is k m. What is their combined length?

Answer. $(20 + k)$ metres. **Why?** We add lengths because the pipes are joined end-to-end.

Set A • Q3: counting currency notes

Question. Find Krithika's total amount for (i) 3 notes of Rs. 100, 5 notes of Rs. 20, 6 notes of Rs. 5; (ii) $6, 4, 3$ notes respectively; (iii) $8, 4, z$; (iv) x, y, z.

Answer. **(i)** $3(100)+5(20)+6(5) = 430$. **(ii)** $6(100)+4(20)+3(5) = 695$. **(iii)** $800+80+5z = 880+5z$. **(iv)** $100x + 20y + 5z$ rupees.

Set A • Q4: flour mill

Question. A mill needs 10 seconds to start and 8 seconds per kilogram. Find the time for y kg.

Answer. **Option (d):** $10+8y$ **seconds.** The 10-second startup happens once, not for each kilogram.

Set A • Q5: phrases into expressions

Question. Write expressions for (a) five more, (b) four less, (c) two less than thirteen times, (d) thirteen less than twice a number.

Answer. Let the number be n: **(a)** $n + 5$, **(b)** $n - 4$, **(c)** $13n - 2$, **(d)** $2n - 13$.

Set A • Q6: invent a story

Question. Describe situations for $8x + 3y$ and $15j - 2k$.

Answer. $8x + 3y$**:** buy eight items costing Rs. x each and three items costing Rs. y each. $15j - 2k$**:** produce 15 objects daily for j days and discard two objects daily for k days.

Set A • Q7: calendar 2-by-3 grid

Question. The bottom-middle date is w. Write the expressions for the other five dates.

Answer. A row of dates increases by 1 to the right; a date one row above is 7 less:

$w - 8$	$w - 7$	$w - 6$
$w - 1$	w	$w + 1$

Figure it Out — NCERT pp. 93–94

Set B • Q1: picture sums

Question. Simplify the totals represented by three coloured number patterns on p. 94.

Answer. **Pattern 1**: $5y - 6 + x + x + 2 + 5y = \boxed{10y + 2x - 4}$. **Pattern 2**: the entries consist of four $2p$, four $3q$, two 3 and two -2: $\boxed{8p + 12q + 2}$. **Pattern 3**: four $-5g$ entries and twelve $5k$ entries: $\boxed{60k - 20g}$.

Set B • Q2: simplify every expression

Question. Simplify (a)–(f) from the p. 94 list, keeping each comma-separated expression in order.

Answer.

(a)	$4p;\ 3p + q;\ 2p$
(b)	$2p - 2q;\ 2q$
(c)	$0;\ -2q$
(d)	$-d;\ -c$
(e)	$c;\ 2d - c$
(f)	$d - 2c$

Why? Add coefficients of *like* terms; a minus sign before brackets changes every sign inside.

Set B • Error hunt, items 1–6

Question. The textbook shows deliberately incorrect simplifications for items 1–6. Write the correct answers.

Answer.

(1) $3a + 2b$ (unlike terms)
(2) 0 for $3b - 2b - b$
(3) $6p + 12$ for $6(p + 2)$
(4) $x - y$ for $(4x + 3y) - (3x + 4y)$
(5) $3 + 6z$ for $5 - (2 - 6z)$
(6) $x + 5$ for $2 + (x + 3)$

Set B • Error hunt, items 7–11

Question. Complete the remaining textbook mistake corrections.

Answer.

(7) $5y - 6$ for $2y + (3y - 6)$
(8) $6p + 3q$ for $7p - p + 5q - 2q$
(9) $30w + 15x$ for $5(2w + 3x + 4w)$
(10) $3j + 6k + 9h + 12$ (already simplest; the factored version is equal but not expanded)
(11) 0 for $4(2r + 3s + 5) - 20 - 8r - 12s$

Pattern noticed: after like terms are combined, each different variable can appear as at most one variable term, plus one constant term.

Figure it Out — NCERT pp. 102–105

Set C • Q1: roti and pulao

Question. Jowar roti is Rs. 30 and pulao Rs. 20 per plate. Find the earnings from x roti plates and y pulao plates.

Answer. **Option (a),** $30x + 20y$ **rupees.** Multiply each price by its own quantity.

Set C • Q2: flags

Question. Pushpita gives one tiny flag to each of p champak-only, q marigold-only, and r both-flower customers.

Answer. **Option (a), $p+q+r$ flags.** Customers buying both types receive one flag, not two.

Set C • Q3: snail

Question. A snail climbs u cm by day and slips d cm each night for 10 days and 10 nights. What is net displacement? What if $d > u$?

Answer. $\boxed{10(u-d)}$ cm. If $d > u$, this displacement is negative, so the snail is **below its starting position** after 10 cycles (subject to the well's physical bottom).

Set C • Q4: Radha cycles

Question. Daily distance is 5 km in week 1, then increases by z km in each subsequent week. Total distance in three seven-day weeks?

Answer. $7(5) + 7(5+z) + 7(5+2z) = \boxed{105+21z}$ km.

Set C • Q5: flowchart

Question. Start with $w+2$ and fill operation-path blanks in the diagram on p. 102.

Answer. Shown paths give: $w+2-5 = w-3$, then multiplying by 3 gives $3w-9$; and $w+2-8 = w-6$, then subtracting 4 gives $w-10$. The upper-right path is $w+2+3 = w+5$, then $\times 4$ gives $4w+20$. **Notice:** the printed bottom-right branch shows $3w-6$ alongside $+3$ then $\times 4$, which does not agree algebraically for arbitrary w; that branch cannot satisfy all printed labels simultaneously. Check with your teacher instead of forcing an inconsistent answer.

Set C • Q6: train journey

Question. A train travels from Yahapur to Vahapur with three intermediate stations. Each section takes t minutes and stops are 2 minutes. (a) $t = 4$? (b) expression?

Answer. There are **four moving sections** and **three stops**: $4t + 3(2) = \boxed{4t+6}$ minutes. For $t = 4$: $16 + 6 = \boxed{22 \text{ minutes}}$.

Set C • Q7: simplify six expressions

Question. Work out parts (a)–(f), simplifying the brackets and collecting like terms.

Answer.

(a)	$4a+5b+10$	(b)	$a-13b-16$
(c)	$12x+6$	(d)	$6x+15$
(e)	$h+4$	(f)	$21m-20n+5$

Example: (a) $3a+9b-6+8a-4b-7a+16 = (3+8-7)a+(9-4)b+(-6+16)$.

Set C • Q8: addition of expressions

Question. Find the sums of all six pairs in textbook Q8 (p. 103).

Answer.

(a)	$13d+c-2$	(b)	$7f+4s-4$
(c)	$2c-d-2$	(d)	$-7f-4s+3$
(e)	0	(f)	0

Example: (a) $(4d-7c+9)+(8c-11+9d) = 13d+c-2$.

Set C • Q9: subtract expressions

Question. Subtract each given first expression *from* the second expression (all six parts).

Answer.

(a)	$-3a+15b-32$	(b)	$18x+16y-23$
(c)	$-27g+10h+2$	(d)	$-3a-3b-32$
(e)	$-13x-13y+6$	(f)	$-16g+3h+30$

Example: (a) $(6a+9b-18)-(9a-6b+14)=-3a+15b-32$.

Set C • Q10: invent situations

Question. Give stories that fit $8x+3y$ and $15x-2x$.

Answer. $8x+3y$**:** eight mangoes at Rs. x each plus three bananas at Rs. y each. $15x-2x$**:** of fifteen items priced Rs. x each, two were returned; the remaining sales value is $13x$.

Set C • Q11: rope folds

Question. A straight rope cut once gives 2 pieces. Fold once then cut once and get 3. What happens after 10 folds, or r folds (as shown in the figure)?

Answer. The pictured repeated-fold pattern increases the number of pieces by one per fold: at 10 folds, 12 **pieces**; at r folds, $r+2$ **pieces**. This follows the specific fold-and-cut arrangement in the diagram, not every imaginable fold.

Set C • Q12: squares of matchsticks

Question. The pictured row adds one side-sharing square at each step. How many matchsticks are needed for 10 squares and w squares?

Answer. First square requires 4; every new square shares one side and adds 3. Thus $4+3(10-1)=\boxed{31}$. General formula $4+3(w-1)=\boxed{3w+1}$.

Set C • Q13: traffic-light colours

Question. Colours follow red, yellow, green, yellow, then repeat. Find colours at 90, 190, 343; describe positions.

Answer. **90: yellow; 190: yellow; 343: green.** Red positions are $4n-3$; green are $4n-1$; yellow are $2n$, with $n=1,2,\ldots$.

Set C • Q14: growing X-shape

Question. The number of squares in successive pictured steps is $5,9,13,\ldots$. Find steps 4, 10, 50, a general formula and total vertices.

Answer. Each step adds 4 squares, so $S_n=5+4(n-1)=4n+1$. Thus $S_4=17$, $S_{10}=41$, $S_{50}=201$. As the small squares shown have distinct vertices, the count is $4S_n=\boxed{16n+4}$.

Set C • Q15: endless four-column grid

Question. (a) Formula by column; (b) locate 124, 147, 201; (c) formula for row r, column c; (d) spot patterns of multiples of 3.

Answer. **(a)** Column c is $4(r-1)+c$, for $c=1,2,3,4$; specifically $4r-3, 4r-2, 4r-1, 4r$. **(b)** 124 is row 31, col 4; 147 is row 37, col 3; 201 is row 51, col 1. **(c)** $N=4(r-1)+c$. **(d)** Multiples of 3 shift one column left from one row to the next (wrapping around); column 4 contains multiples of 4 and even numbers appear in columns 2 and 4.

In-text examples, puzzles and discussion checks

Age relationship • pp. 81–82

Question. Shabnam is three years older than Aftab. Express their ages and evaluate examples.

Answer. Using a for Aftab and s for Shabnam: $s = a + 3$ and $a = s - 3$. If $a = 23$, $s = 26$; if $s = 20$, $a = 17$.

Matchstick L shapes • p. 82

Question. Each L uses two matchsticks. Express the number needed for n Ls.

Answer. $2n$ sticks. Examples: 5 Ls need 10 sticks, 7 Ls need 14.

Coconuts and jaggery • pp. 82–83

Question. Coconuts cost Rs. 35 each and jaggery Rs. 60 per kg. Find costs of (10 coconuts, 5 kg) and (8 coconuts, 9 kg).

Answer. General cost $35c + 60j$. At $(c, j) = (10, 5)$: Rs. 650; at $(8, 9)$: Rs. 820.

Arithmetic refresher • p. 85

Question. Evaluate the seven sample arithmetic expressions in section 4.2.

Answer. In order: $\mathbf{3}, \mathbf{130}, \mathbf{40}, \mathbf{56}, \mathbf{37}, \mathbf{82}, \mathbf{100}$. **Check:** $42 + 15 - (8 - 7) = 56$. $20 + 8(16 - 6) = 100$.

Difference between expressions • p. 93

Question. Are $10y - 3$ and $10(y - 3)$ equal?

Answer. No. Their difference is $10y - 3 - (10y - 30) = 27$. For example at $y = 2$, they are 17 and -10.

Calendar diagonals • pp. 98–99

Question. If the top-left of a 2-by-2 calendar square is a, prove that the diagonal sums agree.

Answer. The four entries are $a, a + 1, a + 7, a + 8$. The diagonal sums are $a + (a + 8) = 2a + 8$ and $(a + 1) + (a + 7) = 2a + 8$. Equal for every a.

Matchstick pattern • pp. 100–101

Question. At step y, there are y horizontal sticks and $y + 1$ diagonal ones. Find total.

Answer. $y + (y + 1) = 2y + 1$. Equivalent repeated-addition rule: $3 + 2(y - 1) = 2y + 1$.

Student notes: learn the ideas

What do letters mean?

A **letter-number** is a symbol like x or n that can stand for a number. An **algebraic expression** combines letter-numbers, ordinary numbers and operations: for instance $3x + 5$ means *three times x, then add 5*. You can **substitute** a particular value to evaluate it.

Key concept

If $x = 4$, then $3x + 5 = 3(4) + 5 = 17$. The letter x is not the multiplication sign, and $3x$ means $3 \times x$.

Visualise a growing pattern



1 square(s) 2 square(s) 3 square(s) 4 square(s)

Adjacent squares need $4, 7, 10, 13, \ldots$ matches: the rule is $3w + 1$. **Explain** the pattern before writing a formula.

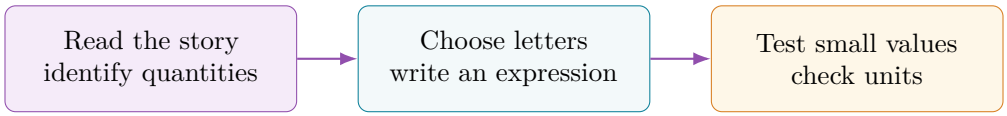
Brackets, like terms and signs

1. Distribute: $4(2x + 3) = 8x + 12$.
2. Combine like terms: $5a + 3b - 2a + b = 3a + 4b$.
3. Subtract a whole bracket: $8x - (2x - 3) = 8x - 2x + 3 = 6x + 3$.

Common mistake

$3x + 2$ is **not** $5x$. x and ordinary numbers are unlike terms. When subtracting $(a - b)$, change **both** signs: $-(a - b) = -a + b$. A one-time fixed cost must not be multiplied by the number of items.

Model a story with a formula



Remember

Use **P–E–C**: **P**attern or problem statement → **E**xpression → **C**heck by substitution. Two formulas are equivalent only if they give the same result for *every* permitted value.

Short extra practice with solutions

1. A notebook costs Rs. n. Cost of 4 notebooks and a Rs. 7 pen? $4n + 7$.
2. Simplify $5(x + 2) - 3x$. $2x + 10$.
3. A pattern starts at 6 and increases by 3 each step. Expression for step n? $6 + 3(n - 1) = 3n + 3$.
4. $x = 5$: calculate $7x - 2(x + 1)$. 23.
5. Is $3(a + 1)$ the same as $3a + 1$? **No:** they differ by 2.

One-page revision checklist

Remember

- I can translate words and figures into correct algebraic expressions.
- I can substitute a value and calculate an expression correctly.
- I can distribute brackets and combine like terms, without mixing unlike terms.
- I can explain $3w + 1$ and $4n + 1$ pattern rules.
- I have worked through the three "Figure it Out" sets, including all 15 final questions.

Source: NCERT Class 7 uploaded textbook. Explanations and diagrams are original educational adaptations. Measurement/drawing questions are to be verified on the textbook figure.